Here's a comprehensive project plan for "Rentro," a professional and production-leve...

Okay, this is an excellent starting point! Based on the comprehensive project plan you've outlined for Rentro, here is a detailed Product Requirements Document (PRD). This document translates the "what" (features) into "how" (user stories, acceptance criteria) and "why" (business goals, user needs).

Product Requirements Document (PRD) - Rentro Platform

Document Version: 1.0
Date: March 28, 2026
Product Manager: [Your Name/Team]
Stakeholders: Engineering Team, Design Team, Marketing Team, Business Development

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1. Introduction

Rentro is envisioned as the leading digital platform connecting individuals seeking Paying Guest (PG) accommodations with property owners and managers across various cities. The platform aims to resolve common pain points in the PG search and listing process by offering a secure, transparent, and efficient marketplace. This PRD outlines the functional and non-functional requirements for the Rentro platform, focusing on the Minimum Viable Product (MVP) and a roadmap for future enhancements, ensuring a professional and production-level system.

2. Goals & Objectives

Overall Product Goal: To become the premier online platform for PG accommodations, simplifying the process for both seekers and owners.

SMART Objectives for MVP Launch:

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- **Acquisition:** Onboard 500 PG owners and 5,000 PG seekers within the first 3 months post-launch in target cities.
- **Engagement:** Achieve a minimum of 20% owner-to-seeker inquiry response rate within 1 month of an owner listing a property.
- **Satisfaction:** Maintain an average user satisfaction score of 4 out of 5 stars (via post-inquiry surveys) from both seekers and owners regarding platform usability and effectiveness within 6 months.
- **Performance:** Ensure an average page load time of under 3 seconds on a 4G connection for key user flows (search, listing view, dashboard).

3. Target Audience & Personas

3.1. PG Seeker Persona: "Ravi the Student"

- **Demographics:** 18-24 years old, college student, new to a city.
- **Needs:** Affordable, safe, gender-specific PG near college, includes food/Wi-Fi, easy communication with owner, quick search.
- **Pain Points:** Difficulty finding PGs online, unclear amenities, high brokerage, unsafe locations, unreliable owner contact.

3.2. PG Owner Persona: "Mrs. Sharma the Proprietor"

- **Demographics:** 45-65 years old, owns multiple PG properties.
- **Needs:** Easy way to list and manage properties, receive direct inquiries, showcase amenities effectively, manage availability, minimize vacancies.
- **Pain Points:** Relying on brokers, manual record-keeping, difficulty reaching wide audience, managing inquiries from multiple channels.

4. Scope (MVP)

In Scope for MVP:

- Secure User Registration & Login (Seeker, Owner)
- Basic User Profiles (Seeker, Owner)
- PG Listing Creation, Editing, Deletion (Owner)
- Detailed PG Listing Display with Photos & Map (Seeker)
- Location-Based Search & Essential Filters (Seeker)
- "Contact Owner" / "Send Inquiry" functionality (Seeker to Owner)
- Owner Dashboard with Inquiry Management

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- Seeker Dashboard with Shortlist & Inquiry Tracking
- Basic Responsive Design
- Admin Panel for User/Listing Moderation
- Core Legal Pages & Help/FAQ

Out of Scope for MVP (Planned for Phase 2+):

- In-app Chat/Real-time Messaging
- Review & Rating System
- Online Booking & Payment Gateway Integration
- AI-powered Recommendations
- Advanced Analytics for Owners
- Virtual Tour Integration
- Multi-language Support
- Advanced User Verification (KYC)

5. Functional Requirements (User Stories & Acceptance Criteria)

5.1. User Management & Authentication

- **US1: Secure Registration**
  - **As a new user (Seeker or Owner),** I want to securely register with my email and password, so I can access the platform's features.
  - **Acceptance Criteria:**
    - User can choose role (Seeker/Owner) during registration.
    - System performs email validation.
    - Password complexity requirements are enforced.
    - Account activation via email verification link.
- **US2: User Login**
  - **As a registered user,** I want to log in using my email and password, so I can access my personalized dashboard.
  - **Acceptance Criteria:**
    - Login form is available on the homepage.
    - Error messages are displayed for incorrect credentials.
    - "Remember Me" option (optional, if implementing).

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- **US3: Forgot Password**
  - **As a registered user who forgot my password,** I want to initiate a password reset process, so I can regain access to my account.
  - **Acceptance Criteria:**
    - User receives a password reset link via email.
    - The link is time-limited and single-use.
- **US4: User Profiles**
  - **As a user (Seeker/Owner),** I want to create and edit my profile details, so I can personalize my experience and provide necessary information.
  - **Acceptance Criteria:**
    - Seeker profile: Name, phone number, city, occupation.
    - Owner profile: Name, phone number, company name (if applicable), address.
    - Users can upload a profile picture.
- **US5: Role-Based Access**
  - **As a platform,** I want to differentiate between user roles (Seeker, Owner, Admin), so I can provide appropriate access and functionality.
  - **Acceptance Criteria:**
    - PG Owner dashboard and features are only accessible to Owners.
    - PG Seeker dashboard and features are only accessible to Seekers.
    - Admin panel is only accessible to authenticated administrators.

5.2. PG Owners Features

- **US6: Add New PG Listing**
  - **As a PG Owner,** I want to easily add a new PG listing, so seekers can discover my property.
  - **Acceptance Criteria:**
    - Form includes fields for:
      - Property Name/Title
      - Detailed Address (street, locality, city, state, pin code)
      - Property Type (Apartment, House, Hostel, etc.)
      - Rent (per bed/room, monthly, currency)

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        - Deposit Amount
        - Notice Period (days)
        - Number of Rooms/Beds available
        - Room Types (Single, Double, Triple Sharing)
        - Gender Preference (Male, Female, Co-ed)
        - Amenities Checklist (Wi-Fi, AC, Food, Laundry, Geyser, Parking, Wardrobe, etc.)
        - Rules & Regulations (text area)
        - Detailed Description of the PG
    - Users can upload multiple high-resolution images.
    - Users can input a video URL (e.g., YouTube) for a virtual tour.
    - Integrated map for location pin (Google Maps API).
    - Ability to mark individual rooms/beds as available/booked.
- **US7: Edit/Delete PG Listing**
    - **As a PG Owner,** I want to edit or delete my existing PG listings, so I can keep information current or remove properties no longer available.
    - **Acceptance Criteria:**
        - Owner can access a list of their active and inactive listings.
        - Owner can modify any field of an existing listing.
        - Owner can permanently delete a listing after confirmation.
- **US8: Owner Dashboard Overview**
    - **As a PG Owner,** I want a dashboard to view my listings and key metrics, so I can manage my properties effectively.
    - **Acceptance Criteria:**
        - Displays a list of all active/inactive PG listings.
        - Shows a count of total inquiries received.
        - Basic metrics: number of listing views, inquiries per listing.
- **US9: Manage Inquiries**
    - **As a PG Owner,** I want to view and respond to inquiries from seekers, so I can connect with potential tenants.
    - **Acceptance Criteria:**

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    - A dedicated section in the dashboard to view incoming inquiries.
    - Each inquiry shows seeker's name, contact (masked initially), and message.
    - Owner can respond via a structured form.
    - Owner's contact details are only revealed upon the owner's discretion or after initial interaction.

5.3. PG Seekers Features

- **US10: Search for PGs**
  - **As a PG Seeker,** I want to search for PGs based on location, so I can find options in my desired area.
  - **Acceptance Criteria:**
    - Search bar supports city, locality, and landmark keywords.
    - Search results are displayed in a list and/or map view.
- **US11: Filter Search Results**
  - **As a PG Seeker,** I want to apply advanced filters to search results, so I can narrow down options that meet my specific criteria.
  - **Acceptance Criteria:**
    - Filters available: Rent range, room type (single, double, etc.), gender preference, amenities (multi-select), food options, sharing capacity, proximity to landmarks.
    - Filters can be applied individually or in combination.
- **US12: View Detailed PG Listing**
  - **As a PG Seeker,** I want to view a comprehensive page for each PG, so I can make an informed decision.
  - **Acceptance Criteria:**
    - Displays all details provided by the owner (rent, amenities, rules, description).
    - Interactive image gallery.
    - Embedded video (if provided).
    - Integrated map showing exact PG location.
    - "Contact Owner" button.
- **US13: Contact PG Owner**
  - **As a PG Seeker,** I want to send an inquiry directly to a PG owner, so I can ask questions or express interest.

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    - **Acceptance Criteria:**
        - A form is presented on the listing page to send a message.
        - Seeker's contact details are sent with the inquiry.
        - Confirmation message after sending inquiry.
- **US14: Shortlist PGs**
    - **As a PG Seeker,** I want to shortlist PGs, so I can easily revisit my preferred options.
    - **Acceptance Criteria:**
        - A "Save" or "Heart" icon on listing cards/detail pages.
        - A dedicated section in the seeker dashboard to view all shortlisted PGs.
- **US15: Seeker Dashboard**
    - **As a PG Seeker,** I want a dashboard to track my inquiries and manage my shortlisted PGs, so I can organize my search.
    - **Acceptance Criteria:**
        - Displays a list of all inquiries sent, with their status (e.g., Pending, Responded).
        - Displays the list of shortlisted PGs.
        - Link to edit seeker profile.
- **US16: Inquiry Notifications**
    - **As a PG Seeker,** I want to receive notifications when an owner responds to my inquiry, so I don't miss important communication.
    - **Acceptance Criteria:**
        - Email notifications are sent when an owner responds.
        - (Optional for MVP) In-app notification icon updates.

5.4. General Platform Features

- **US17: Responsive Design**
    - **As a user,** I want to access the Rentro website seamlessly on any device (desktop, tablet, mobile), so I can use it conveniently from anywhere.
    - **Acceptance Criteria:**
        - UI adjusts gracefully to different screen sizes.
        - All features are fully functional on mobile devices.
- **US18: Admin Panel - User Management**

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    - **As an Administrator,** I want to manage user accounts, so I can maintain platform integrity.
    - **Acceptance Criteria:**
        - Ability to view, activate, deactivate, or delete user accounts.
        - Ability to reset user passwords.
- **US19: Admin Panel - Listing Moderation**
    - **As an Administrator,** I want to moderate PG listings, so I can ensure quality and adherence to guidelines.
    - **Acceptance Criteria:**
        - Ability to view new and existing listings.
        - Ability to approve, reject, or request edits for listings.
        - Ability to edit any listing details.
- **US20: Admin Panel - Content Management**
    - **As an Administrator,** I want to manage static content like FAQ and Legal Pages, so I can keep information current.
    - **Acceptance Criteria:**
        - CMS interface to edit FAQ, Terms & Conditions, Privacy Policy.
- **US21: Help & Support**
    - **As a user,** I want access to help resources, so I can find answers to common questions or contact support.
    - **Acceptance Criteria:**
        - A visible "FAQ" section.
        - A "Contact Us" form for general inquiries.
- **US22: Basic SEO**
    - **As Rentro,** I want my website to be discoverable by search engines, so potential users can find us easily.
    - **Acceptance Criteria:**
        - Proper title tags, meta descriptions for key pages.
        - Semantic HTML structure.
        - XML sitemap generated.

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6. Non-Functional Requirements

- **Performance:**
  - Load time for primary pages (homepage, search results, listing detail) must be under 3 seconds.
  - API response times for critical actions (login, search) under 1 second.
  - Support for at least 1,000 concurrent users at MVP launch without performance degradation.
- **Scalability:**
  - The architecture must be designed to scale vertically and horizontally to accommodate future growth in users and data without significant re-architecture.
- **Security:**
  - HTTPS/SSL enforcement for all traffic.
  - Protection against common web vulnerabilities (XSS, CSRF, SQL Injection).
  - Regular security audits and penetration testing.
  - User data encryption at rest and in transit.
  - Role-based access control rigorously enforced.
- **Reliability:**
  - Target uptime of 99.9% for core services.
  - Robust backup and disaster recovery mechanisms for all data.
- **Usability (UX/UI):**
  - Intuitive and easy-to-navigate interface for both seekers and owners.
  - Clear calls to action.
  - Consistent design language across the platform.
- **Maintainability:**
  - Well-documented code and API endpoints.
  - Modular architecture for easier updates and debugging.
- **Compliance:**
  - Adherence to data privacy regulations (e.g., GDPR principles, local data protection laws).
  - Legal pages (T&C, Privacy Policy) must be legally sound.

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7. Technical Requirements (High-Level)

- **Architecture:** Microservices-oriented (if complexity warrants, otherwise monolithic with clear separation of concerns), cloud-native.
- **Frontend:** React.js / Next.js for SSR and performance. Tailwind CSS for styling.
- **Backend:** Node.js (NestJS) or Python (Django) for API development.
- **Database:** PostgreSQL for relational data, Redis for caching/sessions. Elasticsearch or Algolia for search.
- **Cloud Platform:** AWS/GCP/Azure for hosting, managed services (DB, storage, compute).
- **Version Control:** Git (GitHub/GitLab).
- **CI/CD:** Automated pipelines for testing and deployment.

8. Analytics & Tracking

- **Key Performance Indicators (KPIs):**
  - Number of registered PG Seekers and Owners.
  - Number of active PG listings.
  - Number of inquiries sent/received.
  - Owner response rate to inquiries.
  - User retention rate (monthly active users).
  - Page views and average session duration.
  - Conversion rate (search to inquiry).
- **Tools:** Google Analytics, Mixpanel, or similar analytical tools integrated. Custom dashboards for internal monitoring.

9. Monetization Strategy (Initial)

For MVP, the focus is on user acquisition and engagement. Initial monetization will be indirect or basic:

- **Future:** Premium/Featured Listings for owners (charging for higher visibility). This would require additional features to implement, but the framework should be considered.
- **Future:** Basic Subscription Tiers for Owners (e.g., more listings, limited analytics).

10. Dependencies & Assumptions

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